Maxwell Equations in Covariant Form

Sandro Vitenti

Scope

These notes build on the calculus tools developed previously and show how Maxwell equations naturally unify into a single geometric statement on spacetime.

The key steps are:

- recognizing that $\mathbf{E}$ and $\mathbf{B}$ are observer-dependent projections,
- combining them into the Faraday tensor $F_{\mu\nu}$,
- writing homogeneous and non-homogeneous Maxwell equations covariantly,
- understanding the geometric role of differential forms in this framework.

From 3D Vector Calculus to Covariant Electromagnetism

Motivation

Classical electromagnetism is usually formulated with a preferred time slicing, where fields are split into spatial vectors $\mathbf{E}$ and $\mathbf{B}$. This formulation is not covariant: under Lorentz transformations, time mixes with space, and electric and magnetic fields mix with each other.

This suggests that $\mathbf{E}$ and $\mathbf{B}$ are not fundamental objects. Instead, they should arise as components of a single spacetime object.

Our goal is to rewrite Maxwell equations in a form that:

- does not depend on a preferred frame,
- is manifestly Lorentz covariant,
- has a clear geometric meaning.

Homogeneous vs Non-Homogeneous Equations

Maxwell equations naturally split into two groups:

Non-homogeneous (contain sources):

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \times \mathbf{B} - \frac{1}{c^2}\partial_t \mathbf{E} = \mu_0 \mathbf{J},$$

where ρ is the charge density, $\mathbf{J}$ is the current density, and ε_0, μ_0 are the vacuum permittivity and permeability. Finally, $c = 1/\sqrt{\varepsilon_0 \mu_0}$ is the speed of light.

Homogeneous (purely geometric):

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0.$$

The homogeneous equations do not depend on sources. This suggests they encode a purely geometric structure.

Hidden Metric in Vector Calculus

Operators like gradient, divergence, and curl implicitly use the metric. For instance, the gradient is not just ∂_i, but effectively:

$$(\nabla f)^i = g^{ij}\partial_j f.$$

This becomes important when changing coordinates or working covariantly. The form language makes these dependencies explicit.

From Vectors to Forms

To make the geometry explicit, we reinterpret fields as differential forms:

- Electric field $\mathbf{E}$ becomes a 1-form:
$$E = E_i \, \mathrm{d}x^i,$$
- Magnetic field $\mathbf{B}$ becomes a 2-form:
$$B = \frac{1}{2} B_{ij} \, \mathrm{d}x^i \wedge \mathrm{d}x^j.$$

The magnetic field is naturally a 2-form because it is associated with flux through surfaces, which are 2-dimensional.

Unification into a Single Spacetime Equation

The Key Observation

The homogeneous equations:

- Faraday law: $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$,
- No magnetic monopoles: $\nabla \cdot \mathbf{B} = 0$,

are actually different components of a single spacetime equation.

Spacetime Coordinates

Introduce spacetime coordinates with signature $(-,+,+,+)$:

$$x^0 = ct, \qquad x^i \quad (i = 1,2,3) \text{ spatial.}$$

The Electromagnetic Field Tensor

Define the **Faraday tensor** $F_{\mu\nu}$ as a 2-form on spacetime with components:

$$F_{0i} = -\frac{E_i}{c}, \qquad F_{ij} = B_{ij} = \epsilon_{ijk} B_k,$$

and antisymmetry $F_{\mu\nu} = -F_{\nu\mu}$. This matches the conventions, where the same tensor is fixed by $F^{0i} = E^i/c$, $F^{ij} = \epsilon^{ijk} B_k$.

In matrix form:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix}.$$

The Homogeneous Maxwell Equation

The homogeneous Maxwell equations become:

$$\mathrm{d}F = 0.$$

This is a single equation on spacetime, encoding all four components:

- the 3 spatial components of Faraday's law,
- the 1 component of $\nabla \cdot \mathbf{B} = 0$.

Verification in Components

The equation $\mathrm{d}F = 0$ expands to:

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0.$$

For $(\alpha, \mu, \nu) = (0, i, j)$, using $\partial_0 = \frac{1}{c}\partial_t$:

$$\partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = 0,$$

substituting $F_{ij} = B_{ij}$, $F_{j0} = -F_{0j} = E_j/c$, and $F_{0i} = -E_i/c$, then multiplying through by c, gives:

$$\partial_t B_{ij} + \partial_i E_j - \partial_j E_i = 0,$$

exactly Faraday's law.

For $(\alpha, \mu, \nu) = (i, j, k)$ (cyclic):

$$\partial_i B_{jk} + \partial_j B_{ki} + \partial_k B_{ij} = 0,$$

which is equivalent to $\nabla \cdot \mathbf{B} = 0$.

Geometric Interpretation

Form Degree and Dimension

- F is a 2-form on spacetime (6 independent components),
- $\mathrm{d}F$ is a 3-form on spacetime (4 independent components in 4D).

In 4 dimensions, a 3-form has exactly

$$\binom{4}{3} = 4$$

independent components. These correspond to:

- the 3 components of Faraday's law,
- the 1 component of $\nabla \cdot \mathbf{B} = 0$.

Thus, covariance naturally combines them into a single geometric statement.

Closure Condition

The identity $\mathrm{d}^2 = 0$ guarantees consistency. If F is itself exact, $F = \mathrm{d}A$, then:

$$\mathrm{d}F = \mathrm{d}^2 A = 0$$

automatically. This motivates the introduction of the electromagnetic potential A_μ.

The Non-Homogeneous Equations

Source Term

The non-homogeneous equations involve the current 4-vector:

$$J^\mu = (c\rho, J^x, J^y, J^z).$$

They are written covariantly as:

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu,$$

or equivalently, using the Hodge dual:

$$\mathrm{d} \star F = -\mu_0 \star J.$$

Component Check

For $\nu = 0$, using $F^{i0} = -E^i/c$:

$$\partial_i F^{i0} = -\mu_0 J^0 = -\mu_0 c\rho,$$

i.e. $-\frac{1}{c}\nabla \cdot \mathbf{E} = -\mu_0 c\rho$, which gives Gauss's law $\nabla \cdot \mathbf{E} = \mu_0 c^2 \rho = \rho/\varepsilon_0$.

For $\nu = k$ (spatial), using $\partial_0 = \frac{1}{c}\partial_t$, $F^{0k} = E^k/c$, and $\partial_i F^{ik} = -(\nabla \times \mathbf{B})^k$ (from $F^{ik} = \epsilon^{ikl} B_l$):

$$\partial_0 F^{0k} + \partial_i F^{ik} = -\mu_0 J^k,$$

which expands to:

$$(\nabla \times \mathbf{B})^k - \frac{1}{c^2}\partial_t E^k = \mu_0 J^k,$$

the Ampère-Maxwell law.

Final Structure

Electromagnetism in covariant form:

> **!** Maxwell Equations
>
> **Field strength:**
>
> $$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$
>
> **Homogeneous equations:**
>
> $$\mathrm{d}F = 0.$$
>
> **Non-homogeneous equations:**
>
> $$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu.$$

Conceptual Summary

- The split into **E** and **B** is frame-dependent,
- Vector calculus hides the underlying spacetime geometry,
- Differential forms make the covariant structure explicit,
- Maxwell equations reduce to simple geometric statements on spacetime.

This formulation prepares the ground for:

- understanding gauge freedom ($A_\mu \to A_\mu + \partial_\mu \chi$),
- generalizing to curved spacetime (general relativity),
- interpreting electromagnetism as a $U(1)$ gauge theory.

Exercises

Exercise 1: The Faraday Tensor and the Homogeneous Equation (Essential)

a) Verify that the matrix representation of $F_{\mu\nu}$ given above is antisymmetric and correctly encodes $\mathbf{E}$ and $\mathbf{B}$.

b) Show explicitly that $\mathrm{d}F = 0$ gives the component $\partial_t B_{12} + \partial_1 E_2 - \partial_2 E_1 = 0$, and identify this with one component of Faraday's law.

c) Prove that if $F = \mathrm{d}A$, then $\mathrm{d}F = 0$ automatically.

Exercise 2: The Non-Homogeneous Equation and Gauge Invariance (Essential)

a) Verify that $\partial_\mu F^{\mu 0} = -\mu_0 J^0$ reproduces Gauss's law when $F^{i0} = -E^i/c$ and $J^0 = c\rho$.

b) Show that the gauge transformation $A_\mu \to A_\mu + \partial_\mu \chi$ leaves $F_{\mu\nu}$ unchanged.

Exercise 3: Electrodynamics from an Action Principle (Essential)

Write the action for free electromagnetism,

$$S = -\frac{1}{4\mu_0} \int F_{\mu\nu} F^{\mu\nu} \, \mathrm{d}^4 x,$$

and derive the vacuum Maxwell equations by varying A_μ.

Exercise 4: Dimension-Counting and the Hodge Dual

a) In 2+1 dimensions, how many independent components does $F_{\mu\nu}$ have? How many does $\mathrm{d}F$ have?

b) Explain why the Hodge dual maps a 2-form to another 2-form in 4D specifically, and what role this plays in writing the non-homogeneous equations as $\mathrm{d} \star F = -\mu_0 \star J$.